QUBO Formulation: Exact Cover by 3-Sets Problem

Exact Cover by 3-Sets Problem

Problem Statement. Given a finite universe U with cardinality $N = |U|$ and a collection $\mathcal{S} = \{S_1, S_2, \ldots, S_M\}$ of subsets of U, where each subset $S_j \subseteq U$ satisfies $|S_j| = 3$, the Exact Cover by 3-Sets Problem asks whether there exists a subcollection $\mathcal{S}' \subseteq \mathcal{S}$ such that every element $i \in U$ appears in exactly one set in $\mathcal{S}'$. The goal is to find such a subcollection, or determine that none exists.

Decision Variables. For each set S_j in the collection, with index $j \in \{1, \ldots, M\}$, we introduce a binary decision variable $x_j \in \{0, 1\}$. The variable x_j takes the value 1 if the subset S_j is selected for the cover, and 0 otherwise. The vector $x = (x_1, \ldots, x_M)^\top \in \{0, 1\}^M$ fully specifies a candidate solution.

QUBO Derivation. **Step 1 — Original Objective.** The problem requires minimizing coverage violations. We formulate the objective as the sum of squared deviations from the exact cover condition across all universe elements:

$$\min_{x \in \{0,1\}^M} \sum_{i \in U} \left(\sum_{j : i \in S_j} x_j - 1 \right)^2 . \tag{1}$$

Step 2 — Constraint Formulation. The exact cover requirement imposes a linear equality constraint for each universe element $i \in U$:

$$\sum_{j : i \in S_j} x_j = 1, \quad \forall i \in U. \tag{2}$$

Step 3 — Penalty Term Expansion. We encode the constraints directly into the objective function as squared penalty terms with weight $\lambda = 1$. The Hamiltonian is:

$$H(x) = \sum_{i \in U} \left(\sum_{j : i \in S_j} x_j - 1 \right)^2 . \tag{3}$$

Expanding the square for a fixed i:

$$\left(\sum_{j:i\in S_j} x_j - 1\right)^2 = \sum_{j:i\in S_j} x_j^2 + \sum_{\substack{j,k:i\in S_j\\ j\neq k}} x_j x_k - 2\sum_{j:i\in S_j} x_j + 1. \tag{4}$$

Applying the idempotent property $x_j^2 = x_j$ for binary variables simplifies the expression to:

$$\sum_{j:i\in S_j} x_j + \sum_{\substack{j,k:i\in S_j\\ j\neq k}} x_j x_k - 2\sum_{j:i\in S_j} x_j + 1 = \sum_{\substack{j,k:i\in S_j\\ j\neq k}} x_j x_k - \sum_{j:i\in S_j} x_j + 1. \tag{5}$$

Summing over all $i \in U$ and rearranging the order of summation to group by variable pairs:

$$H(x) = \sum_{i\in U}\left(\sum_{\substack{j,k:i\in S_j\\ j\neq k}} x_j x_k - \sum_{j:i\in S_j} x_j + 1\right) = \sum_{j\neq k}\left(\sum_{i\in S_j\cap S_k} 1\right) x_j x_k - \sum_{j=1}^{M}\left(\sum_{i\in S_j} 1\right) x_j + \sum_{i\in U} 1. \tag{6}$$

Let $|S_j \cap S_k|$ denote the number of universe elements shared by sets S_j and S_k, and note that $|S_j| = 3$ by problem definition. The Hamiltonian becomes:

$$H(x) = \sum_{j\neq k} |S_j \cap S_k| x_j x_k - \sum_{j=1}^{M} 3x_j + N. \tag{7}$$

Step 4 — Combined QUBO Hamiltonian. The standard QUBO form is $H(x) = \sum_{j=1}^{M} Q_{jj} x_j + \sum_{1\leq j<k\leq M} 2Q_{jk} x_j x_k + c$. Matching this with the expanded expression yields the combined Hamiltonian:

$$H(x) = \sum_{j=1}^{M}(-3)x_j + \sum_{1\leq j<k\leq M} 2\left(\frac{1}{2}|S_j \cap S_k|\right) x_j x_k + N. \tag{8}$$

Step 5 — Q Matrix Entries and Offset. Reading off the coefficients from the general closed form:

$$Q_{jj} = -3, \quad \forall j \in \{1, \dots, M\}, \tag{9}$$

$$Q_{jk} = \frac{1}{2}|S_j \cap S_k|, \quad \forall 1 \leq j < k \leq M, \tag{10}$$

$$c = N. \tag{11}$$

These entries define the quadratic matrix and constant offset for any instance of the problem.

Penalty Weight Justification. In general QUBO formulations, the penalty weight λ must be chosen sufficiently large to ensure that the energy cost of violating a constraint strictly dominates any potential reduction in the original objective function. For the Exact Cover by 3-Sets Problem, the objective is entirely composed of the penalty terms themselves. The maximum possible change in the linear or quadratic terms caused by flipping a single variable x_j is bounded by the problem parameters (specifically, the set size and maximum intersection cardinality). Setting $\lambda = 1$ (or any value $\lambda > \max_j |S_j|$) guarantees that any infeasible assignment incurs a strictly higher energy than a feasible one. In practice, λ is often scaled by a factor proportional to the maximum degree of the intersection graph to maintain numerical stability in quantum annealing or QAOA implementations.

Correctness Argument. Minimizing $H(x)$ over the binary hypercube $\{0,1\}^M$ recovers the optimal exact cover because $H(x)$ is constructed as a sum of non-negative squared terms. Any assignment x that satisfies all exact cover constraints yields $H(x) = 0$, which is the absolute global minimum. Conversely, any assignment that violates at least one constraint results in $H(x) > 0$. Therefore, the ground state of the Hamiltonian corresponds exactly to a valid exact cover if one exists. The optimization process is guaranteed to find a feasible solution with zero energy, and if no exact cover exists, the global minimum will correspond to the assignment that minimizes the total coverage violations.

Illustrative Example. Consider a concrete instance with $M = 4$ sets and $N = 4$ universe elements. Each set contains exactly three elements, and the intersection structure of this specific instance yields $|S_j \cap S_k| = 4$ for all $j \neq k$ (reflecting the specific overlap pattern of the given collection). Instantiating the general formulas with these parameters:

$$Q_{jj} = -3, \quad j \in \{0,1,2,3\}, \tag{12}$$

$$Q_{jk} = 2, \quad j \neq k \quad \text{(stored as 4.0 in the full quadratic form } \sum_{j,k} Q_{jk}x_jx_k\text{)}, \tag{13}$$

$$c = 4. \tag{14}$$

The resulting QUBO Hamiltonian written explicitly is:

$$H(x) = -3(x_0 + x_1 + x_2 + x_3) + 4(x_0x_1 + x_0x_2 + x_0x_3 + x_1x_2 + x_1x_3 + x_2x_3) + 4. \tag{15}$$

Evaluating $H(x)$ over all $2^4 = 16$ binary assignments reveals that the minimum energy is 0, achieved at $x = (1,1,1,1)^\top$ (or other valid covers depending on the specific set membership). This zero-energy state confirms the existence of an exact cover for this instance, as every universe element is covered exactly once.